Operating Systems · FCIM / FAF · UTM · 2026-2027

The File System with Your Own Hands

Lab 5 · Track A: warm-up on Linux · 90 minutes

Goal

Everything from today's lecture, checked on your own Linux laptop: the tree and the file types, inodes behind the names, hard and symbolic links, permissions, and the disks underneath. No programming today: commands, careful observation, and short explanations.

Before you start

- Work as your normal user, NOT as root (root ignores most permission checks and the experiments would lie).

```
whoami                  # must not print root
mkdir -p ~/os-lab5 && cd ~/os-lab5
```

Part 1. The tree and the file types (~15 min)

```
ls /
ls -ld /etc /etc/passwd /bin /dev/null /tmp
ls -l /dev | head
file /usr/bin/ls /etc/passwd /bin
du -sh /var/log 2>/dev/null ; du -sh /usr 2>/dev/null
```

Observe (write in report.md)

(1) For each path in the second line, write its type from the first letter of ls -l. (2) What is /bin on your system: a directory or a link, and to where? (3) Find one character device and one block device in /dev. (4) Which is bigger on your laptop, /var/log or /usr, and by roughly how much?

Part 2. Inodes and links (~25 min)

1. An inode, and a name that is not part of it:

```
echo hello > notes.txt
stat notes.txt
ls -li notes.txt
```

2. Hard link, then remove the "original":

```
ln notes.txt hard.txt
ls -li notes.txt hard.txt
rm notes.txt
cat hard.txt ; ls -li hard.txt
```

3. Symbolic link, then remove its target:

```
ln -s hard.txt soft.txt
ls -li hard.txt soft.txt
cat soft.txt
rm hard.txt
cat soft.txt ; ls -l soft.txt
```

4. Directories have links too. Predict the link count first:

```
mkdir d ; ls -ld d
mkdir d/x d/y ; ls -ld d
```

5. Size is not space:

```
echo x > one.txt ; ls -l one.txt ; du -h one.txt
truncate -s 1G big.img ; ls -lh big.img ; du -h big.img
rm big.img
```

Observe (write in report.md)

(1) Which number in stat is the inode, and which tells how many names the file has? (2) After rm notes.txt, why could you still read hard.txt? What does rm really remove? (3) Why did cat soft.txt fail at the end, although soft.txt still exists? How big is soft.txt and why exactly that size? (4) A new directory has 2 links, and 4 after you create two subdirectories. Explain where each link comes from. (5) Why does a 2-byte file use 4.0K, and a 1 GB file use 0?

Part 3. Permissions (~25 min)

6. Files:

```
echo secret > s.txt
chmod 600 s.txt ; ls -l s.txt
chmod 000 s.txt ; cat s.txt
chmod u+r s.txt ; cat s.txt
umask ; touch new.txt ; mkdir newdir ; ls -ld new.txt newdir
```

7. Directories: predict each result before you run it.

```
mkdir box ; echo hi > box/f
chmod u=r,go= box  ; ls box ; cat box/f     # read only
chmod u=x,go= box  ; ls box ; cat box/f     # execute only
chmod 755 box
```

8. Special bits and other people's files:

```
ls -l /usr/bin/passwd ; ls -ld /tmp
cat /etc/shadow
ls -l /etc/shadow
```

Observe (write in report.md)

(1) Write 600, 644, 755 and 700 as rwx strings, and give one real use for each. (2) With chmod 000 you own the file but cannot read it. Why does ownership not help? (3) What is your umask,

and how does it explain the permissions of new.txt and newdir? (4) Explain the box experiment: what does r allow on a directory, and what does x allow? (5) What do the s in /usr/bin/passwd and the t in /tmp mean, and why does passwd need the s?

Part 4. Disks and the kernel as files (~15 min)

```
lsblk
df -h
mount | grep " / "
grep -c processor /proc/cpuinfo
head -3 /proc/meminfo
head -5 /proc/self/status
ls -l /proc/self/fd
```

Observe (write in report.md)

(1) Draw your disk: disk name, partitions, which one is mounted at / and with what file system type. (2) How full is your root file system? (3) How many CPU cores and how much RAM does your laptop have, according to /proc? (4) /proc/self/status showed the name of a program. Which one, and why that one?

Stretch (optional)

Pick any two and add them to report.md. (a) Magic numbers: gzip -c /etc/passwd > notes.txt ; file notes.txt ; xxd notes.txt | head -1. Which bytes tell file the real type? (b) Hunt for setuid programs: find / -xdev -perm -4000 -type f 2>/dev/null. Is there anything on your laptop that is not on the lecture slide? (c) Delete a file that is still open: exec 3< bigfile ; rm bigfile ; ls -l /proc/$$/fd. What does the link say, and when is the space really freed? (close it with exec 3<&-) (d) With sudo: truncate -s 64M disk.img ; mkfs.ext4 disk.img ; sudo mount -o loop disk.img /mnt ; df -h /mnt ; df -i /mnt ; sudo umount /mnt. How many inodes did mkfs create?

Wrap-up and deliverable (~10 min)

You submit:

- Your lab report (see below), with the answers to all "Observe" boxes and the outputs that support them.

Lab report (required):

- Every lab must be accompanied by a written report: what you did, the answers to all "Observe" boxes with the commands and outputs (or screenshots) that support them, and a short conclusion.
- Labs 5 and 6 may be combined into a single report. If you do this, give each lab its own clearly labelled section and submit the report once, together with the Lab 6 code.
- Everything you submit (the report and any code) must also be pushed to your Git repository.